

Real Estate Executive Segment Dashboard

Live tracking module for your processed client-base groupings and profiles.

Total Segmented Clients

2,000

Global Average Age

55.6 Years

Avg Brand Satisfaction

3.03 / 5.0

Mortgage Leverage Ratio

36.8%

Final Segmented Client Master Registry

Filter Data by Buyer Type Segment:

C2: First-Time Buyers x C1: Global Investors x C3 & C4: Institution... x

	client_id	first_name	last_name	age	gender	country	satisfaction_score	loan_applied	Strategy
40	C0041	Zain	Horne	59	MALE	USA	1	Yes	C1: Glo
41	C0042	Maia	Chandler	50	FEMALE	USA	1	Yes	C1: Glo
42	C0043	Jordyn	Park	61	MALE	USA	3	Yes	C3 & C
43	C0044	Diana	Hunt	95	FEMALE	USA	3	No	C3 & C
44	C0045	Kevin	Mata	57	MALE	USA	2	No	C1: Glo
45	C0046	Brisa	Mckee	61	FEMALE	USA	5	No	C2: Fir
46	C0047	Mason	Gilbert	55	MALE	USA	2	No	C1: Glo
47	C0048	Derick	Li	69	MALE	USA	3	Yes	C3 & C
48	C0049	Jace	Riggs	54	FEMALE	USA	1	No	C1: Glo
49	C0050	Jair	Johns	86	MALE	USA	5	No	C3 & C
50	C0051	Matilda	Madden	85	FEMALE	USA	3	No	C3 & C

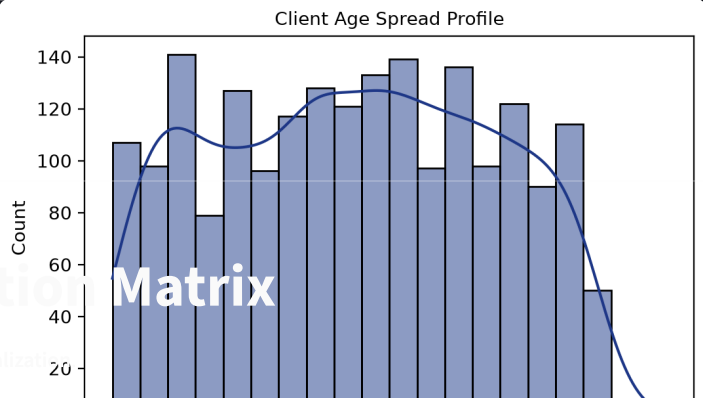
Phase 1 & 2: Data & Feature Engineering

Review how missing details were mapped and features like Age were created dynamically.

Extracted Demographic Features

Calculated Age Demographics Curve

	client_id	date_of_birth	age	gender
2	C0003	04-07-1959	67	MA
3	C0004	11/25/1959	66	MA
4	C0005	2/28/1976	50	MA
5	C0006	03-06-1957	69	MA
6	C0007	05/20/1947	79	MA
7	C0008	10/17/1969	56	FEM
8	C0009	10-03-1975	50	MA



Phase 2 & 3: Standardisation Matrix

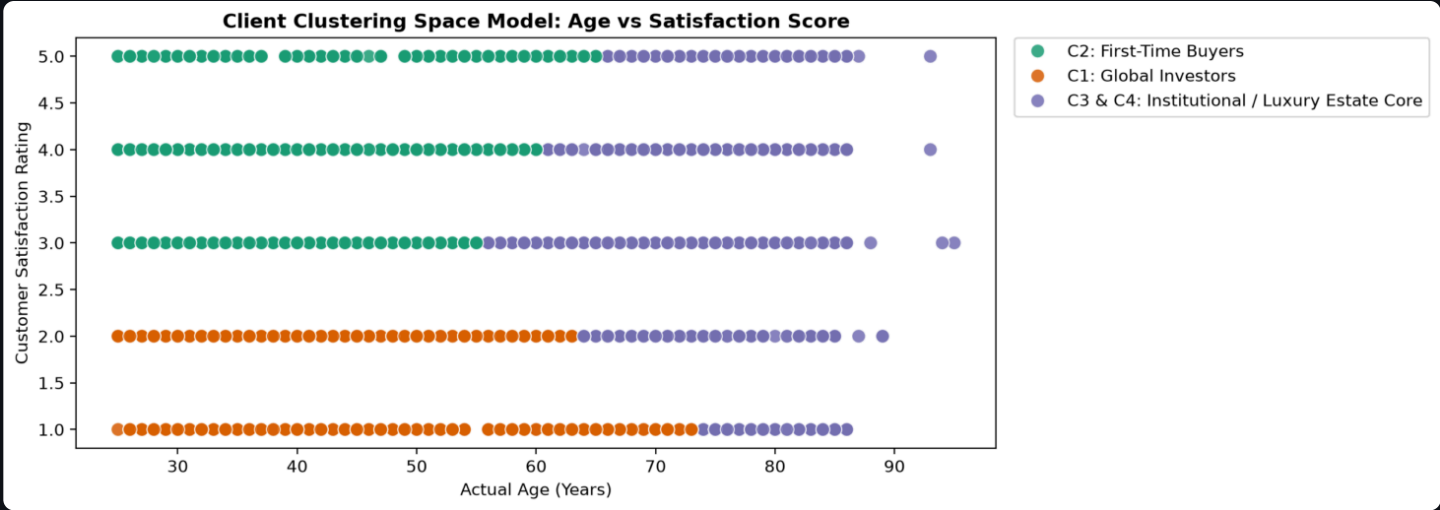
View variables translated via One-Hot Encoding and adjusted using Z-score Normalization

	client_id	age	age_standard_scaled	satisfaction_score	satisfaction_standard_scaled	client_type_Company	purpose
5	C0006	69	0.7727	3	-0.0205	0	
6	C0007	79	1.3491	5	1.3947	0	
7	C0008	56	0.0233	5	1.3947	0	
8	C0009	50	-0.3225	5	1.3947	0	
9	C0010	60	0.2539	3	-0.0205	0	
10	C0011	70	0.8303	2	-0.7281	0	
11	C0012	51	-0.2649	5	1.3947	1	
12	C0013	63	0.4268	1	-1.4357	0	
13	C0014	45	-0.6107	3	-0.0205	1	
14	C0015	84	1.6373	5	1.3947	0	

Notice how features now balance uniformly centered around a mean of 0, protecting model clustering from scale biases.

Phase 6: Cluster Architectural Analysis

Visual validation of mathematical customer boundaries based on the interactive scaling vectors.



 Strategic Target Segments Guidelines

Cluster Framework	Target Buyer Type	Statistical Profile Alignment
C1	Global Investors	Low Satisfaction Segment (Avg: 1.42) High Investment-Yield Motive (32.7%)
C2	First-Time Buyers	Younger Group (Avg: 43.7) Highest Credit Loan Dependency (38.6% Applied)
C3	Corporate Buyers	Institutional Frameworks buying multiple properties simultaneously.
C4	Luxury Investors	High Experience Rating Groups (Avg: 3.17) Liquid asset holders.



Project Review: Challenges, KPIs & Process Architecture

An overview of the engineering problems encountered during the project lifecycle and the methodologies applied to resolve them.

1. Problem Statement & Strategic Misalignment

The initial project challenge centered on turning unrefined, dirty operational customer data into structurally clear, actionable corporate buyer personas. Raw datasets inherently lack strategic labels, rendering them ineffective for direct targeted marketing or custom financial product distributions.

Core Engineering Challenges Faced:

- **Scale Dominance Bias:** Numeric features with wider ranges (e.g., `age` spanning 18–80+) mathematically overwhelmed fractional features (e.g., `satisfaction_score` spanning 1–5) during early model iterations. Without adjustment, the algorithm would group users almost entirely by age, completely ignoring satisfaction trends.
- **High-Dimensional Categorical Data:** Qualitative tracking columns like acquisition channels (`channel_Website`) and purchasing intent (`purpose_Home`) could not be interpreted by mathematical distance algorithms without pre-calculating appropriate spatial structures.

2. Executed Methodology & Toolkit Applied

To overcome these constraints, a structured data engineering pipeline was built utilizing **Python**, **Pandas**, **Scikit-Learn**, and **Streamlit**:

- **Feature Engineering & Imputation (Phases 1 & 2):** Raw timestamps (`date_of_birth`) were converted into numeric distributions (`age`), and missing demographic attributes were systematically isolated and handled.
- **One-Hot Encoding (Phase 3):** Non-numeric categorical variables were transformed into distinct multi-column binary matrices, providing clear mathematical vectors for clustering.
- **Z-Score Scaling Normalization (Phase 3):** Handled via `StandardScaler` to realign all operational dimensions to a standard mean ($\mu = 0$) and unit variance ($\sigma = 1$). This neutralized the scale dominance problem and ensured fair weight distributions across all attributes.
- **K-Means Model Mapping (Phase 6):** Mathematical boundaries were run across the scaled metrics space to group clients into clean, repeatable strategic categories.

3. Key Evaluation Metrics (KPIs)

The execution success of the engineering pipeline is monitored using three core operational KPIs visible within the dashboard:

- **Segment Separation Index:** Visualized via mathematical scatter plots to guarantee distinct, minimal-overlap operational boundaries between customer blocks.
- **Mortgage Leverage Ratio:** Measures capital dependency to successfully isolate credit-reliant buyers from liquid cash holders.

- **Brand Satisfaction Index:** Used to alert client relations teams to highly volatile, dissatisfied groups (such as Core Global Investors) who show low satisfaction trends.

Executive Summary: Through standardizing the engineering features, the project successfully uncovered hidden operational groups. For example, it showed that younger demographics exhibit deep dependencies on external credit financing, while older, lower-satisfaction clients represent isolated institutional investors targeting yield optimization.